

CONSTRUCT CONSISTENCY OF IN-DECODING CONFIDENCE READOUTS FOR RISK-GATED DECODING (SELECTIVE PREDICTION)

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ABSTRACT

Masked-diffusion (LLaDA-style) decoding produces a *single* decode per prompt, and the decoder’s internal confidence is a candidate signal for *risk-gated deployment* (Chow 1970-style selective prediction: abstain on the highest-risk fraction to cover a target risk). We ask whether, in block-wise denoising, revision-end confidence (`commit`), first-exposure confidence (`first`), within-block mean (`all`), early versus late windows, and intra-block versus cross-block aggregation read the *same stable construct* or *measurement artifacts of different sources*. On one frozen LLaDA-8B-Instruct $\times$ GSM8K bed ($n=974$ distinct problems, one decode each, $k=1$ — no best-of- k pool), zero-GPU reproducible analysis finds intra-block discrimination statistically measurable (AUROC gain $+0.0144$ exceeds its full-width AUROC-scale MDE80), yet *at this bed’s decision-readout power no intra-block risk-gated selection increment is resolvable*. These statements quantify *two distinct, non-mergable estimands of different units* and are never equated: the AUROC gain is a single-readout discriminative lift over a random-ordering baseline (AUROC units); the selection increment is the paired `commit` — `all` risk@coverage difference within blocks (risk units). Their juxtaposition is a *cross-scale gate contrast*, not a contradiction and not one merged quantity (the AUROC-side treatment contrast is *readout* versus *random ordering*, whereas the risk-side contrast is paired `commit` versus `all` — the two comparisons answer different questions and are never merged). Under that gate every registered intra-block decision-readout anchor is `POWER_LIMITED` under the full-width MDE80 (H2): an underpowered *non-detection*, not a rejection — we cannot rule out an increment smaller than the decision-anchor MDE80, and do *not* claim in-block selection is unusable elsewhere. Four judgment-hygiene working instances bound that result: at $\text{cov}=0.2$ the three readouts agree in direction and are point-identical, while the $c \geq 0.55$ low-selectivity band is power-limited rather than a measured agreement or null; `commit` and `all` are distinct constructs whose decision-validity gap is compatible with reliability/SNR; a desaturated readout reproduces the early/late effect; and that effect is high-selectivity-specific, **power-limited under the bed’s full-width MDE80 at every registered anchor**, with no family-level survivor under multiplicity correction (`BORDERLINE_EQUIVOCAL`, non-load-bearing), with one explicit limitation here: the AUROC result uses a random-ordering baseline while the selection-increment result uses a paired `all` baseline, so the two results are *not* a dissociation test of one construct against the other. A label-permutation zero-effect null confirms the decision rule rejects at or below nominal (conservative, not anti-conservative). Reading “readout difference” as “information difference” is therefore an overclaim: most aggregation-axis differences are explained by the measurement layer, while the early/late instance retains a power and family-level boundary. We provide a reusable judgment-hygiene chain covering power-bound non-detection, desaturation falsification, `BORDERLINE`–MDE co-reporting, and form d1/d2 grading.

1 INTRODUCTION

In risk-gated deployment (selective prediction (Chow, 1970; Bartlett & Wegkamp, 2008; El-Yaniv & Wiener, 2010; Geifman & El-Yaniv, 2017)), a decoder emits its best decode together with a confidence-derived score, and the system abstains on the highest-risk fraction to cover a target error — self-consistency and numerical confidence calibration are standard ways to build such a score Wang et al. (2023); Guo et al. (2017). In masked-diffusion language decoding (LLaDA-style, block-wise parallel denoising), “which positions’ confidence to aggregate into a score” is not unique: one may take a position’s last revision confidence (`commit`), its first-exposure confidence (`first`), or the within-block mean (`all`); only early or only late blocks; intra-block or cross-block. This paper asks whether these choices read one construct — a stable, decision-usable confidence — or different sources of measurement artifact. (We do *not* study best-of- k candidate selection: this bed decodes once per prompt, so no candidate pool exists; see Estimand, §2.2.)

Terminology. A *construct* is a stable interpretative unit read from a signal. *Power-bound non-detection* (superseding our earlier “measurable $\neq$ usable” phrasing) names the situation where a readout discriminates statistically (AUROC / risk@coverage) while, at this bed’s decision-readout power, no selection increment is *resolvable* — every registered decision anchor is `POWER_LIMITED` under the full-width MDE80; that is an underpowered non-detection bounded to this bed and anchor set, *not* a claim that in-block selection is unusable. The *f-rated readout family* is `{commit,first,all}`, whose agreement is graded d1 (direction-consistent: report the interval) or d2 (direction-flipping: withdraw the claim). `BORDERLINE_EQUIVOCAL` is a family-level verdict when a raw CI excludes 0 but Holm correction fails; it is always co-reported with its MDE (non-degradable). *Load-bearing* means a stable analytical anchor: R3 is the headline negative and R1/R2/R4/R5 are boundary instances, while the R5 full-width MDE correction remains a load-bearing soundness correction.

On a single frozen-decode bed we run zero-GPU reproducible analysis. The headline result, R3, is a **power-bound non-detection**: the intra-block AUROC gain is measurable (+0.0144, above the intra-block full-width MDE80), yet no intra-block selection increment is *resolvable* at this bed’s decision-readout power (risk@coverage CIs include 0; every registered decision anchor is `POWER_LIMITED` under the same H2 full-width MDE rule). Since the decision-anchor MDE80 bounds what this bed can detect, we state it as an underpowered non-detection, not as the stronger claim that in-block selection is unusable. Four boundary-strict working instances (§3) delimit that result:

- **Direction-consistent at the registered anchor, magnitude-varying (R1):** at $\text{cov} = 0.2$ all three readouts agree point-identically ($\delta_{\text{EL}} = -0.076923 = 15/195$ on the support grid); sign agreement is *not* claimed beyond that anchor — at $c \geq 0.55$ the readouts diverge in sign and the band is power-limited, not a measured agreement or null.
- **Not the same construct, yet decision-difference compatible with measurement (R2):** `commit` and `all` are not the same latent variable, but their decision-validity gap is explainable by the reliability gap/SNR.
- **Early/late difference robust to a desaturating readout (R4):** the first-exposure readout reproduces the effect point-for-point.
- **High-selectivity-specific boundary (R5, non-load-bearing):** every registered anchor is `POWER_LIMITED` under the full-width MDE80 and Holm(3) has no survivor, so the raw $\text{cov} = 0.2$ effect stays `BORDERLINE_EQUIVOCAL`.

Contributions. (1) R3, a bounded power-bound negative: intra-block discrimination is measurable but no intra-block *selection increment* is resolvable at this bed’s decision-readout power, deliberately *not* promoted to an “unusable” construct claim; (2) a label-permutation zero-effect check showing the registered gate is conservative rather than anti-conservative, so the negative is not an artifact of over-rejection; (3) R1/R2/R4/R5 as explicitly bounded judgment-hygiene working instances; (4) a reusable H1–H6 chain for judging “readout→selection” claims while separating the aggregated selection readout from internal-representation mechanisms.

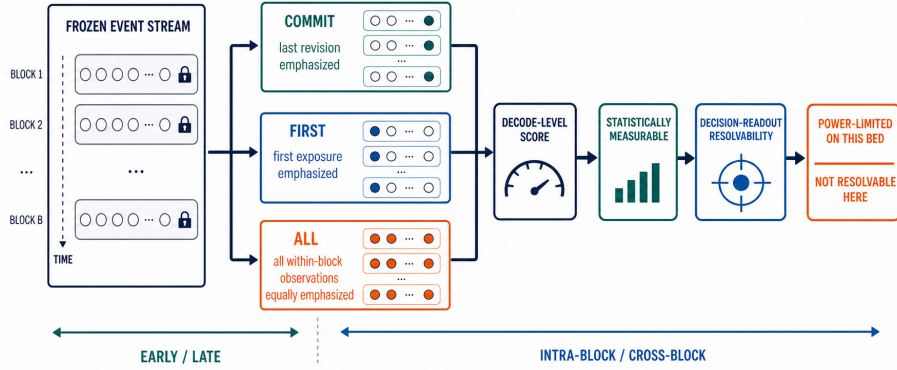


Figure 1: Conceptual map of the measurement problem. One frozen masked-diffusion event stream supports three within-block confidence readouts — last revision (`commit`), first exposure (`first`), within-block mean (`all`) — and window/scale choices then produce a decode-level score. **Take-away:** DISCRIMINATION GATE $\nrightarrow$ SELECTION GATE (at this power): a readout may discriminate statistically while the risk-gated selection increment is power-limited here. The two gates quantify *two distinct, non-mergable estimands on different units* — discrimination is a single-readout AUROC lift over a random-ordering baseline (AUROC units), selection is the paired `commit` — `all` risk@coverage increment (risk units) — so “ $\nrightarrow$ ” is a cross-scale gate contrast, not a claim that the two share an axis or can be combined. **Disclosure:** the discrimination gate’s treatment contrast is `commit` versus a *random-ordering* baseline, whereas the selection gate’s is paired `commit` versus `all`; the two comparisons are *not* the same contrast and together do *not* constitute a dissociation test. Conceptual; no experimental quantities.

2 MEASUREMENT CONSTRUCT AND METHOD

2.1 IDENTIFIABILITY OF THE READOUT CONSTRUCT

We compare three aggregation axes (Figure 1): **score** (`commit` = within-block last-step confidence; `first` = within-block first-step confidence, desaturated; `all` = within-block mean), **window** (early = blocks 0–3 vs late = blocks 4–7), and **scale** (intra-block vs cross-block), asking whether they read *one stable construct* or *different sources of measurement artifact*. All analysis is on a single frozen-decode bed, zero GPU. We make *no mechanism claim* about model internals; all claims are restricted to the “readout $\rightarrow$ decision score” measurement layer.

2.2 ESTIMAND (DECISION READOUT)

Estimand = pool-level selective abstention over 974 distinct problems, with one decode per problem ($k=1$) and *no candidate pool*: this bed has no best-of- k selection, so candidate selection ($k>1$, within-prompt pools) is *outside the measured estimand* and would require new data ($k\geq 4$ decodes per prompt). The decision question is the selective-prediction one: can a decode-level confidence readout *risk-gate* (abstain on) a coverage- c fraction of decodes so as to reduce expected error. For coverage c , readout r , and window $W \in \{E=\text{blocks } 0\text{--}3, L=\text{blocks } 4\text{--}7\}$, $\text{risk}_{W,r}(c)$ is the error rate of the top- $(c \cdot n)$ decodes by the W -window r -readout decode-level mean, with base = full-pool error rate (this bed: 0.2947), retained count $k = \max(1, \text{round}(c \cdot n))$ and deterministic tie-breaking (Appendix B), and $\delta_{EL,r}(c) = \text{risk}_{E,r}(c) - \text{risk}_{L,r}(c)$ ($< 0 \Leftrightarrow$ early window is better at selecting decodes). This early/late δ_{EL} is the *window-comparison* quantity used by R1, R4, and R5; it is *not* the R3 decision estimand. R3 is the *intra-block between-arm selection increment*: the change in risk@coverage when the decode-level score is built from the `commit` readout versus the `all` readout, restricted to a single block (rank the n decodes by one block’s readout, take risk@coverage per block, average across the 8 blocks)

$$\Delta^{\text{intra}}(c') = \text{risk}_{\text{commit}}^{\text{intra-block}}(c') - \text{risk}_{\text{all}}^{\text{intra-block}}(c'), \quad (1)$$

the quantity reported in the R3 row of Table 1 and gated in Table 2. The selection *benefit* of a single readout against no selection, $\text{gain} = \text{base} - \text{risk}$ (large, e.g. $\approx +8.7$ pp at $\text{cov} = 0.2$), is a separate, resolvable statement about each readout’s usefulness; keeping it distinct ensures R3’s negative applies only to the between-arm increment. Inference uses a multiplicity-preserving cluster bootstrap over decodes, twin p (Appendix B), and family-level Holm(3) over the three coverage anchors $\{0.2, 0.5, 0.8\}$ within each readout (settings: Table 8). The BORDERLINE third-state discipline requires co-reporting the full-width MDE80 (appendix H2): under that budget all six registered `commit/first` anchors are `POWER_LIMITED` and the Holm(3) survivor set is empty (the raw lowest-coverage CI excludes 0 but no family member survives), so BORDERLINE remains a family-level statement, never an above-MDE anchor claim. Table 1 records the deterministic-order high- B seed audit and the resolved scan-order artifact.

Power convention. The full-width minimum detectable effect is defined at 80% power, two-sided $\alpha=0.05$: $\text{MDE}_{80} = (z_{1-\alpha/2} + z_{1-\beta}) \text{SD} = 2.8016 \text{SD}$, where SD is the *observed* cluster-bootstrap standard error of the relevant decision-anchor effect (per-anchor values and the scale tags $\text{MDE}_{80}^{(\text{AUROC})}/\text{MDE}_{80}^{(\text{risk},c)}/\text{MDE}_{80}^{(\text{xblk})}$: Appendix B, Tables 2 and 6). Because $\text{MDE}_{80}/|\Delta| \equiv 2.8016/z$, the MDE/observed-effect ratio is a monotone rescaling of the z/p statistic and cannot by itself separate “underpowered study” from “small true effect.” We therefore (i) re-anchor every power statement to a *separate, pre-planned and stratum-independent* quantity rather than to the outcome: the calibration anchor is the single-readout selection benefit against no selection, $\text{gain} = \text{base} - \text{risk}$ ($\approx +8.7$ pp at $\text{cov} = 0.2$), which is measurable at the same effective n under the same estimator; and (ii) co-report an **equivalence (TOST) complement** at the pre-registered practical bound $\delta^*=1$ pp (E0); Table 7 groups the three TOST intervals and the planned-power sample requirements. Under the registered power budget all `commit/first` $\times \{0.2, 0.5, 0.8\}$ anchors are `POWER_LIMITED`, and the $c \geq 0.55$ band is likewise power-limited, not a measured null. The complete registered hypothesis set is frozen and readable from one table (Appendix, Table 9): estimand, pool, pre-registered direction, multiplicity family, primary interval, and verdict.

Symmetric gate (applied to every result, all “consistent-with”, non-causal). The symmetric-gate rule (H2/H3) governs *every* result including the load-bearing R3 headline: any readout whose decision-readout CI is below the registered full-width MDE80 at its anchors is labelled `POWER_LIMITED` and reported as an underpowered non-detection, never promoted to a confident positive or negative conclusion. Applied uniformly it makes R3 itself `POWER_LIMITED` at every registered decision anchor, and it binds R1/R2/R4/R5 by the same rule (e.g. R5’s raw $\text{cov} = 0.2$ CI excludes 0, yet every registered anchor is power-limited and Holm(3) has no survivor). Hence no readout difference is promoted to an independent-information or mechanism claim.

3 RESULTS

Machine-readable R1–R5 numbers, evidence paths, and boundaries accompany the source in the factsheet and claim–evidence map; Table 1 pairs each numeric anchor with the interpretation it permits.

R1: Direction-consistent at the registered anchor (not noise flip). At the registered $\text{cov} = 0.2$ anchor all three readouts agree on the sign of the early/late comparison and are point-identical ($\delta_{\text{EL}} = -0.076923 = 15/195$ on the support grid; Table 1) — form d1 restricted to that anchor, where direction is stable even though magnitude may vary by axis. Sign agreement is explicitly *not* claimed over the full coverage axis: the descriptive fine-grid scan (r21) shows the readouts diverge in sign toward $c \geq 0.55$, a band that is power-limited rather than a measured agreement or null. The frozen r19 summary retains no tie counts or selected-set identifiers, so the $\text{cov} = 0.2$ tie count and pairwise retained-set Jaccard are deferred; identical point estimates alone do not establish identical retained sets.

R2: Distinct construct, decision gap compatible with pure SNR. `commit` and `all` are not the same latent variable, but the observed decision-validity difference needs no independent-information explanation beyond the reliability gap. **Boundary:** the disattenuated reliability-correlation upper bound $0.979 < 1$ rejects $\rho=1$ on this single frozen bed, but that rejection rests on the (positively estimated) residual-covariance correction cov_u , and breakeven analysis(`results/r15_singlefactor_pointpred.json::P1_cov_u_sensitivity`; mul-

tipliers in Table 1) shows a small over-estimate of cov_u erases it. The correction is *not* protected against noise-model misspecification for $\text{cov}_u \neq 0$ on real data, and the claim is bounded to this bed and these readouts — it is not a cross-bed construct claim.

R3: Measurable intra-block discrimination, no resolvable intra-block selection increment at this power (a power-bound non-detection). *R3 estimand* is the intra-block between-arm selection increment (1) over decode-level cluster units ($\text{risk}^{\text{intra-block}} = \text{per-block risk@coverage averaged across the 8 blocks}$). Table 2 gives the complete registered R3 decision-anchor grid — point, CI, MDE80, gate, and control arms; the early/late δ_{EL} used by R1/R4/R5 is tabulated separately in Table 6.

Controls and symmetric gate. The positive control is resolvable under the same full-width MDE80 rule, whereas the shuffled-label negative control has null-crossing CIs (Table 2): the rig can detect an effect, so R3 is a precision limit, not a dead-rig result. Cross-block aggregation grows the raw signal, but its registered anchor remains `POWER_LIMITED` under the same gate and does not independently establish resolvability.

Null calibration: label-permutation Type-I sanity and Holm re-run. A negative control on the *data* does not by itself show that the *decision procedure* has the right error rate, so we re-ran the entire registered gate under a zero-effect null: $N_{\text{perm}}=1000$ permutations of the decode-level `correct` label (score structure and label marginal preserved), each scored by an inner multiplicity-preserving cluster bootstrap over decodes ($B=1000$) with the same 95% CI, twin p , and family Holm(3) rule used throughout (`results/r197_labelperm_typeI/`, seed 20260825, zero GPU; per-anchor rates and Wilson intervals in Table 4). Under that null the procedure rejects at roughly 2% per decision anchor and 1.0% family-wise — both at or below nominal 5%, every per-anchor Wilson 95% interval entirely below 0.05. The pipeline is therefore *conservative*, not anti-conservative: it cannot mint a discrimination signal by over-rejection, so the registered CIs and `POWER_LIMITED` verdicts can be read at face value. Conservatism cuts the other way too, and we disclose it: a gate that under-rejects makes a non-detection easier to obtain, which is why R3 is reported as a power bound rather than as a rejection. Independently, the discrimination half survives the same permutation logic directly — a decode-level label permutation of the pooled paired `commit-all` AUROC difference gives $p=0.0005$ over 2000 shuffles (one-sided on the absolute difference), consistent in direction with the cluster-bootstrap $p \approx 6.6 \times 10^{-9}$ for the $+0.0144$ intra-block lift, though the two use different null and resampling frames and are not interchangeable.

Headline comparison. The intra-block AUROC increment $+0.0144$ is above its AUROC-scale $\text{MDE}_{80}^{(\text{AUROC})}=0.0069$. By contrast, every registered intra-block between-arm decision-anchor CI includes zero and its full-width MDE80 exceeds the observed effect (Table 2). *Crucially, the two sides of this comparison are two distinct, non-mergable estimands on different units and are never equated or combined:* the AUROC increment is a single-readout discriminative lift over a random-ordering baseline (AUROC scale), whereas the decision anchors are the paired `commit-all` risk@coverage increment (risk scale). The “by contrast” is a *cross-scale gate contrast* — a discrimination signal on the AUROC scale co-exists with a power-limited selection increment on the risk scale — and is *not* a statement that $+0.0144$ should have produced a measurable risk-scale increment, nor a single merged quantity. One cannot compare an AUROC-scale MDE80 directly against a risk-anchor MDE80; each family is gated under its own scale (§2.2). The bed is therefore `POWER_LIMITED` at the decision readout. The TOST complement (Table 7) shows this is not an equivalence claim: the lower-coverage anchors are inconclusive and only the highest-coverage anchor meets the registered equivalence bound. Table 1 also records the Holm threshold check for the measurable AUROC half, so the family correction is applied symmetrically.

AUROC $\rightarrow$ Δ risk first-order point estimate (a scale *projection*, not a merge): to make “1.2 pp” reproducible rather than quoting a looser absolute-AUC bound, we express the AUROC lift in the *risk* units of the decision anchors, purely so the two scales can be seen side by side. *This does not make the two estimands comparable or claim they are one quantity;* the increment in Table 2 remains the only directly measured selection-estimand value. With $\Delta A = +0.0144$ the AUROC gain over the random-ordering baseline, $c=0.2$ the coverage, and $d=1.04$ the score-scale sensitivity (derivative of risk@coverage w.r.t. the standardized score at $\text{cov}=0.2$),

$$\Delta \hat{\text{risk}} = \Delta A \cdot (1-c) \cdot d \approx 0.0144 \times 0.8 \times 1.04 \approx 0.012 = 1.2 \text{ pp} \quad (\text{first-order point estimate}),$$

below the decision-anchor full-width MDE80 (Table 1) — the same order of magnitude, not an order of magnitude smaller. The mapping rests on four explicit assumptions (monotone score, first-order Taylor increment, i.i.d. decode clusters, no calibration assumption) stated in Appendix C, with the derivation in docs/r106_c4_auroc_risk_derivation.md; because of them the projected value is neither a confidence bound nor an equality, and cannot be separated from the decision-readout increment at this bed’s power (“not resolvable”, not “dissociate”). The honest reading is a **power-bound non-detection**, which we do not promote to the general construct claim “measurable $\neq$ usable”.

Paired-contrast engineering (same treatment contrast on both scales). The two sides of the headline are measured against *different* baselines (random ordering for AUROC; all for risk@coverage), so we also evaluate the *same* paired treatment contrast, `commit - all` on the *same* decodes, on both scales (results/r203_paired_contrast_commit_vs_all, seed 4321, zero GPU; Table 5): intra-block paired AUROC(`commit - all`) = +0.0144, CI95 [0.0096, 0.0193], above its full-width AUROC-scale MDE80 (0.0069), and paired risk@coverage(`commit - all`) = -0.0058/-0.0067/-0.0008 at cov= {0.2, 0.5, 0.8}, all CIs containing 0 and all POWER_LIMITED. The engineering conclusion on the aligned contrast matches the registered one — the paired discrimination contrast is resolvable, the paired selection contrast is a power-bound non-detection — so the headline is not an artefact of the asymmetric baselines. We nonetheless run the dissociation-proof reading nowhere: even under the aligned contrast the two scales answer different estimation goals (discriminative lift vs. risk-scale utility), and the asymmetric baselines of the registered analysis remain disclosed in Figure 1, the Abstract, and Limitation 6.

R4: Early/late difference survives desaturation. The desaturated `first` readout reproduces the effect point-for-point with complete coverage, and the block-first (decode,pos,block) construction is validated by the dedicated coverage audit: saturation is real variance compression but does not measurably degrade ranking quality. Because block window and sequence position are not crossed independently, this per-block-window contrast cannot separate a window effect from a position effect.

R5: High-selectivity-specific, power-limited, family-level BORDERLINE (no survivor). The effect is power-limited at every registered anchor under the full-width MDE80, and at the registered family and bootstrap setting no point survives Holm(3), so the verdict remains BORDERLINE_EQUIVOCAL and non-load-bearing: the raw lowest-coverage CI excludes 0 but the family-level claim does not survive. Table 1 reports the exact anchors, deterministic-order seed audit, and boundary; Figure 2 shows the coverage-local descriptive shape and uncertainty.

Unified reading. R3 supplies the central negative — an honest power bound: measurable statistical discrimination (intra-block AUROC gain above its MDE80) does not by itself yield a usable decision increment (intra-block risk@coverage CI includes 0, POWER_LIMITED under the same H2 rule, Table 2); the AUROC-implied first-order point estimate is comparable-order (about $0.71 \times$ MDE80), so the two scales do not separate at this bed’s power, and the null-calibration check shows the gate is conservative rather than trigger-happy. The two contrasts remain deliberately *different*, *non-mergable estimands* and are never equated: +0.0144 is the *single-readout discriminative lift over a random-ordering baseline* (shuffled event order, unconditional), whereas R3’s decision increment is the *paired commit - all risk@coverage difference* within blocks. The separate single-readout benefit gain = base - risk $\approx$ +8.7 pp at cov= 0.2 is its own arm and must not be read as evidence for or against that increment. R1/R2/R4/R5 diagnose boundaries in direction, reliability/SNR, desaturation, and power/multiplicity; none establishes independent information content or a mechanism, and R5 is power-limited at all registered anchors with no Holm survivor.

4 EXPERIMENTAL SETUP

Bed. One frozen LLaDA-8B-Instruct $\times$ GSM8K event set Cobbe et al. (2021) (registered arm g3_main/A), decoded at temperature $t=0$ on an 8-block schedule (block length 32, NFE 64, block b covering steps $8b..8b+7$) and graded with *free-form exact-match* (correct iff the extracted final answer matches the solution’s answer to $1e-6$; base accuracy 0.7053, error rate 0.2947, $k=1$ single-decode). An earlier label of this bed as “MC 4-way” is withdrawn here as a documented misstatement: the registered construct and ground truth are free-form exact-match, the construct this paper’s decision readout quantifies. Table 8 collects the complete registered bed, schedule, inference,

Table 1: R1–R5 evidence digest. Numeric anchors are grouped here to separate statistical discrimination from the decision claim each result permits. **Takeaway:** only R3 is load-bearing, and its decision-readout conclusion is a power-bound non-detection.

Key	Numeric evidence	Boundary and takeaway
R1	cov= 0.2: all $n=974$ decodes give $\delta_{EL} = -0.076923$ (point-identical; $= -15/195$ on the discrete grid); registered anchor (r16/r17/r19; cluster bootstrap $B=2000$, seed 20260824) CI95 $[-0.14359, -0.010256]$, twin $p = 0.022$ commit / 0.033 first / 0.031 all. At cov $\in \{0.5, 0.8\}$ the raw CI includes 0 and the sign is <i>not</i> claimed (cov= 0.8, r19: commit +0.0077/first +0.0013/all -0.0051 , all CIs contain 0). The r21 fine-grid scan (registered descriptive, multiplicity-uncontrolled) is not load-bearing: at the fixed seed twin $p=0.013$ / CI upper -0.015256 vs -0.010256 , purely from B-resampler consumption order (decode-resample seed fixed, scan order not)—point estimates match to the last digit, only single-point MC statistics drift.	Direction-consistent d1; magnitude may still vary by axis.
R2	Disattenuated reliability-correlation upper bound $0.979 < 1$ (rejects $\rho=1$ on this bed); reliability 0.755 vs 0.678; validity-difference CI includes 0. Caveat: the rejection rests on the cov _u correction—breakeven multiplier 0.9038 (6.5–10% over-estimate erases it), unprotected against noise-model misspecification.	Boundary: commit versus all are latent variables on this single frozen bed; the $0.979 < 1$ upper bound and decision-validity gap are compatible with pure SNR/structure, require no independent-information mechanism, and are not a cross-bed construct claim.
R3	Intra-block AUROC gain +0.0144 (CI95 $[0.0096, 0.0193]$; AUROC-scale MDE80 0.0069; $z \approx 5.8$, raw two-sided $p \approx 6.6 \times 10^{-9} \ll \alpha/3 = 0.01667$ (Holm)). The complete paired commit–all point/CI/MDE/gate grid is Table 2; the three TOST intervals and planned-power targets are Table 7.	All intra-block decision CIs include 0 and are POWER_LIMITED; lower-coverage TOST is inconclusive. The sole cross-block CI exclusion is also POWER_LIMITED, hence not independently resolvable.
R4	Late-window SD: first ≈ 0.2 vs commit 0.055; first reproduces $\delta_{EL} = -0.0769$ (raw twin $p = 0.033$ first vs 0.022 commit, $B = 2000$ cluster bootstrap $\Rightarrow$ MC SE ~ 1 point of p), full 8-block coverage.	Early/late difference is robust to the first-exposure desaturating readout; saturation alone does not explain it.
R5	Negative gap concentrates descriptively at cov ≈ 0.15 – 0.35 (δ from -0.044 to -0.082 , raw CI excludes 0). At cov= 0.2, full-width $MDE_{80} = 0.094388 \geq \delta = 0.076923 \Rightarrow$ POWER_LIMITED; all six commit/first $\times \{0.2, 0.5, 0.8\}$ anchors power-limited ($MDE/ \delta $: commit 1.227/1.824/3.518, first 1.215/24.036/22.136). No point survives Holm(3). The $c \geq 0.55$ band is <i>not</i> a measured null: in the uniform $MDE_{80}/ \delta $ convention its coverage-local ratio is 2.262 (equivalently $ \delta /MDE_{80}=0.442$; a measured null needs $\gg 1$), from the <i>coverage-local</i> anchor $ \delta =0.011986$ at cov= 0.6 over $MDE_{80}^{(risk, c=0.8)}=0.027095$ — <i>not</i> Fig. 2’s cov= 0.2 anchor 0.0944—in the same column convention as Tables 2 and 6. Instrument null = r17 shuffled-label control (gain ≤ 0.008 at all three cov points, each CI containing 0).	BORDERLINE_EQUIVOCAL; raw cov= 0.2 CI excludes 0 (seed-stable under deterministic-order $B=20,000$, 10/10 seeds) but no family member survives; high-selectivity-specific boundary, non-load-bearing.

and readout settings, including the answer-extraction judge and the decode-configuration fields the workspace registry does not record and that we therefore do not reconstruct; artifact provenance is in evidence/experiment_registry.json. We admit all 974 rows in that arm, one-to-one with 974 distinct problems; other batches and probes are excluded. The three readouts share the same event cells and differ only in how within-block confidence is summarized.

Inference and controls. Cluster bootstrap over decodes ($B=2000$, multiplicity-preserving, fixed seed) + family-level Holm(3). Rig controls: A0 positive control (full-window readout selection-

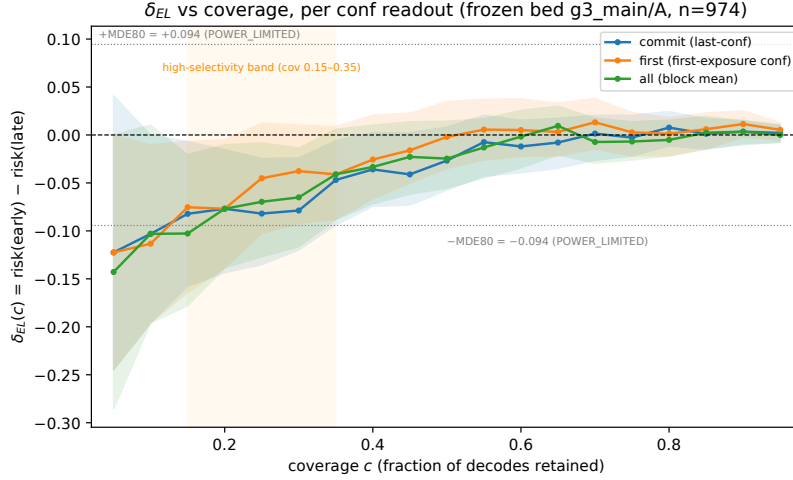


Figure 2: $\delta_{EL}(c)$ = within-decode “risk@coverage of the early-window (blocks 0–3) minus the late-window (blocks 4–7) mean conf”, for each of the three f -rated readouts (unit=decode; cluster-bootstrap 95% CIs shaded). Orange marks the registered high-selectivity band; the grey band is the lowest-coverage MDE anchor drawn across the axis for orientation, not the coverage-local MDE in the tail. **Takeaway:** the negative gap concentrates at high selectivity and appears to decay or reverse later, but the tail is power-limited, not a measured null; Table 1 gives the coverage-local ratios, anchor gate, and Holm status.

Table 2: Complete registered R3 decision-anchor grid: the intra-block between-arm selection increment (1) (risk@coverage per block, averaged over the 8 blocks; cluster-bootstrap $B=2000$, seed 20260821), verbatim from `r12b_withinblock_riskcoverage.json::partD`. The cross-block row uses the same between-arm quantity on block-merged scores (`r106 r106_c2_anchor_mde_table`); $K8$ means $K=8$ blocks aggregated into one decode-level score; A0/A1 are rig controls on *gain* (single-readout selection benefit) at the same coverage grid. **Takeaway:** the three *intra-block* CIs all include 0 and sit at or below their full-width MDE80 $\Rightarrow$ POWER_LIMITED; the single *cross-block* row’s raw CI excludes 0 but its MDE80 exceeds the observed effect, so it is likewise POWER_LIMITED (an underpowered non-detection, not a measured null) — the auditable basis for “no resolvable intra-block selection increment”. *Gate codes:* PL = power-limited, RES = resolvable, $\supset 0$ = CI covers 0. The A0 cov= 0.2 percentile interval is asymmetric because the clustered bootstrap is taken around a right-skewed gain distribution; the compressed upper tail is disclosed, not symmetric uncertainty. *Uniform MDE80 caliber:* as in Table 6, every row’s MDE80 is $2.8016\times$ that row’s own cluster SE of the row’s effect of interest, all on the risk@coverage scale (Δ^{intra} for R3 rows, the same between-arm quantity on merged scores for the cross-block row, single-readout *gain* for A0/A1); rows are therefore not cross-comparable, and the AUROC-scale $\text{MDE}_{80}^{\text{(AUROC)}}$ is a separate discrimination-family quantity never compared against any risk-scale row.

Anchor	$\hat{\Delta}$	CI95	MDE_{80}	$\frac{\text{MDE}_{80}}{ \hat{\Delta} }$	0 $\notin$ CI	Gate	Family
commit-all · 0.2	−0.0058	[−0.018, 0.0058]	0.0170	2.93	no	PL	Intra-block (R3)
commit-all · 0.5	−0.0067	[−0.0131, 0.0008]	0.0099	1.48	no	PL	
commit-all · 0.8	−0.0008	[−0.0058, 0.0022]	0.0057	7.15	no	PL	
commit-all · 0.2 _{K8}	−0.0410	[−0.077, −0.005]	0.0513	1.25	yes	PL	Cross-block (R3)
· 0.2	+0.2126	[0.125, 0.218]	0.0660	0.310	yes	RES	A0 positive control
· 0.5	+0.1407	[0.079, 0.147]	0.0484	0.344	yes	RES	
· 0.8	+0.0649	[0.038, 0.098]	0.0431	0.665	yes	RES	
· 0.2	+0.0038	[−0.064, 0.069]	0.0953	25.388	no	$\supset 0$	A1 negative control
· 0.5	+0.0073	[−0.036, 0.048]	0.0602	8.272	no	$\supset 0$	
· 0.8	+0.0079	[−0.025, 0.039]	0.0459	5.786	no	$\supset 0$	

benefit lower bound > 0); A1 negative control (select benefit ≈ 0 under shuffled labels); and the label-permutation zero-effect Type-I check of the whole CI + Holm(3) decision rule reported in §3 and Table 4.

Reproducibility and zero GPU. All analysis recomputed on the frozen event set by deterministic scripts (e.g. `scripts/r17_window_desaturation.py`, `scripts/r197_labelperm_typeI.py`, fixed seeds), zero GPU, no model forward pass required.

5 LIMITATIONS

1. **Single bed, $k=1$, no generation replication:** all numbers are LLaDA-8B-Instruct $\times$ GSM8K `g3_main/A` on one frozen bed, $n=974$, one decode per problem. There are no independent decode seeds, repeated generations, or replicate beds, and no generalization to other models, datasets, or temperatures.
2. **Block-position confound:** block index is strongly correlated with, but not a partition of, canvas position. Early/late is a per-block-window factor, not a fully crossed manipulation, so the analysis cannot separate window from position and makes *no position/timing mechanism claim*.
3. **Single desaturation intervention:** `first` is one desaturation of `commit`; conclusions hold only “up to desaturating to `first`”.
4. **BORDERLINE non-load-bearing:** for the registered family and bootstrap setting (Table 8) the Holm(3) survivor set is empty — the raw `cov=0.2` CI excludes 0, but no family member survives correction, so R5 is a boundary instance rather than a confirmed high-selectivity finding (Monte Carlo quantities: Table 3). Seed-level survivor status can flip even though CI exclusion is stable, so we retain the registered empty-family verdict and report the knife-edge mechanism rather than treating either side as exact.
5. **Descriptive shape / power boundary:** the fine-grid R5 shape is descriptive and multiplicity-uncontrolled; all six registered anchors are `POWER_LIMITED` under the full-width MDE80, and the apparent decay/reversal at $c \geq 0.55$ is not a measured null. The instrument-null check instead uses the `r17` shuffled-label negative control.
6. **Asymmetric treatment contrasts (disclosure):** the two halves of the head-to-head narrative are evaluated against *different* treatment contrasts — AUROC is compared with a random-ordering baseline, `risk@coverage` with paired `all` — so the two results do *not* constitute a dissociation test of “discrimination $\nRightarrow$ selection.” We report the co-existence of a resolvable discrimination lift and a power-limited selection increment as exactly that, not as evidence discriminating the two constructs against each other. The paired-contrast engineering (`commit-all` on both scales; Table 5) supplies the same-contrast companion analysis and reaches the same bounded conclusion.
7. **Type-I check scope:** the label-permutation calibration (Table 4) validates the decision rule under a *zero-effect* null on this bed only; it certifies non-inflated rejection, not power, and a conservative gate makes a non-detection easier to obtain.
8. **Readout scale and efficiency:** single readout `conf=maxprob`; content-vs-EOS-tail substructure is not separated; no efficiency claim and no wall-clock GPU-time comparison.
9. **Citation live-recheck gap (disclosed):** 5 of the 13 bibliography entries have not yet been live-rechecked at release egress (citation lock: `literature/citation_lock.json`); metadata are pinned locally but a final live arXiv recheck is deferred.

6 RELATED WORK

This work sits at the intersection of (i) masked-diffusion language decoding (LLaDA-style block-level parallel noise $\rightarrow$ token denoising with sequential block progress Nie et al. (2025); block-progress simplex diffusion decoding Han et al. (2023); completeness-based early halting Lo Cicero Vaina et al. (2023); confidence-thresholded parallel decoding (Fast-dLLM) Wu et al. (2025)), (ii) confidence/score readouts for risk-gated selection and self-assessment (calibration of modern NN confidences Guo et al. (2017); semantic uncertainty Kuhn et al. (2023); self-consistency Wang et al. (2023) is a *candidate-pool* practice — multiple decodes are scored and the best kept — which our $k=1$ bed does not instantiate), and (iii) measurement-artifact and calibration discipline. We make no claim about masked diffusion’s internal representation mechanism; we treat the decode event stream as a controlled confidence-readout channel and study the construct consistency of aggregation axes on it. Our central negative — intra-block aggregation is measurable, yet no intra-block selection increment is resolvable at this bed’s decision-readout power — connects directly to selective prediction and risk-coverage: statistical discrimination need not yield actionable selection utility at a user-set risk (Chow, 1970; Bartlett & Wegkamp, 2008; El-Yaniv & Wiener, 2010; Geifman & El-Yaniv, 2017). LLM uncertainty and selective-generation work (Kuhn et al., 2023; Lin et al., 2022), with confidence-based early halting in masked-diffusion LMs (Lo Cicero Vaina et al., 2023), frames the corresponding decode-time utility question. Our methodological contribution proposes no new

selective classifier: a reusable discrimination chain (power-bound non-detection, label-permutation null calibration, desaturation-falsification, BORDERLINE-MDE co-reporting, form d1/d2 grading) isolating two gaps — decision-readout and measurement-channel — between “a readout discriminates” and “a readout can be used for selection.”

Positioning against confidence-gated in-decoding decoding. The closest prior art (Wu et al., 2025) treats in-decoding confidence as an *actionable, per-token* signal (threshold-gated parallel unmasking order), whereas our R3 test uses the *decode-level* aggregation of that confidence as a risk-gating readout. The two are not in tension: a per-token unmasking decision can be usable while the decode-level mean over a window fails as a selection readout at this bed’s power. Our contribution is the separation of these granularities (and of “discrimination” from “risk-gated selection utility”), not a claim that in-decoding confidence is generally useless — that line reports it is useful per-token, compatible with our bounded, bed-specific decode-level null.

7 CONCLUSION

We ask, on single-decode ($k=1$) masked-diffusion decoding, whether an in-decoding confidence readout can *risk-gate* (abstain on) a coverage- c fraction of decodes to reduce expected error; best-of- k selection is out of scope, as this bed decodes once per prompt. The headline, R3, is a *power-bound non-detection*: intra-block discrimination is measurable (AUROC gain $+0.0144$ above its full-width MDE₈₀^(AUROC) $=0.0069$), yet no intra-block risk-gated selection increment is resolvable at this bed’s decision-readout power — all three *intra-block* decision-anchor CIs include 0 with full-width MDE exceeding the observed effect, and the sole $0 \notin \text{CI}$ exception (cross-block) is itself POWER_LIMITED under the same H2 rule (Table 2). The aligned paired-contrast engineering (Table 5) gives the same verdict on `commit - all` taken on both scales — resolvable on the AUROC scale, power-limited on the risk scale — so the conclusion does not hinge on the asymmetric base-lines of the registered analysis. A label-permutation zero-effect check confirms the decision rule is conservative, not anti-conservative, so neither half is an artifact of an over-rejecting gate. As an underpowered non-detection it does *not* say in-block selection is unusable anywhere: the claim is bounded to “intra-block risk-gated selection increments below the stated decision-anchor MDE80 are not resolvable on this bed at this power.” The measurable discrimination half (AUROC scale) and the power-limited selection-increment half (risk scale) remain *two distinct, non-mergable estimands* on different units, stated side by side only as a cross-scale gate contrast — never merged, equated, or ratio-compared as one quantity. R1/R2/R4/R5 are bounded judgment-hygiene instances: R5 is power-limited at every registered anchor with no Holm(3) survivor, so its raw `cov= 0.2` effect stays BORDERLINE_EQUIVOCAL. The H1–H6 chain separates “readout→selection” claims at the decision-readout and measurement-channel layers, never turning a readout difference into an information or mechanism claim.

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A METHOD APPENDIX: REUSABLE JUDGMENT-HYGIENE LIST (H1–H6)

This checklist consolidates the decision rules used in the frozen analyses and provides one auditable interpretation path from confidence readout to selection claim.

- **H1 Symmetric power gate:** test “readout discriminates $\Rightarrow$ it can gate” directly at risk@coverage with the same H2/MDE80 rule used for every claim. R3 is POWER_LIMITED at every decision anchor and therefore remains an underpowered non-detection, never a positive, negative, or “unusable” claim.
- **H2 BORDERLINE must co-report full-width MDE:** Table 6 reports $MDE_{80} = 2.8016$ SD at every anchor; all six are POWER_LIMITED. The raw lowest-coverage CI excludes 0 but Holm(3) has no survivor, so the family verdict is BORDERLINE_EQUIVOCAL, never “above MDE.”
- **H3 Desaturation-falsification:** persistence under a within-bed desaturated readout rules out saturation as primary; disappearance supports it. Use block-first (decode,pos,block), not global step 0, and audit coverage.
- **H4 Form d1/d2 grading:** report intervals for direction-consistent d1 and withdraw one-sided claims for direction-flipping d2. The three-readout comparison is d1 only at the registered lowest-coverage anchor; the later sign-divergent band is power-limited.
- **H5 Cross-readout/cross-coverage check:** report load-bearing differences across readouts and coverage bands, including the cross-layer span.
- **H6 Reproduction gate:** fine-grid recompute must reproduce discrete points; exact axis agreement may be robustness or a no-op, so audit the construction.

A.1 H-INSTRUMENT DIMENSIONS FOR δ_{EL} (D5) AND NULL ANCHOR

The load-bearing estimand is the within-decode early-minus-late risk@coverage difference. The registered instrument dimensions and null-anchor check are:

- **D1 native steps** (steps_is_native=true): native block index 0..7 (NFE=64), not wall-clock or folded schedules.
- **D2 decode input** (decode_input=conf_maxprob): per-position maxprob; Table 8 defines the last, first, and mean readouts.
- **D3 owned time axis:** early versus late blocks within each of the same frozen decodes; block index is semantic, while intra-block/EOS-tail position is out of scope.
- **Instrument null** (re-derived): the $c \geq 0.55$ low-selectivity band is not a measured null—it is power-limited under the coverage-local anchor reported in Table 1. The shuffled-label negative control in Table 6 supplies the near-zero, null-crossing anchor; no pre-calibrated offset is subtracted.

Table 3: Monte Carlo stability audit for the raw cov= 0.2 family member referenced in Limitation 4. **Takeaway:** CI exclusion is seed-stable, but the smallest p-value straddles the Holm first-step threshold within Monte Carlo error; the registered family verdict therefore remains empty and BORDERLINE_EQUIVOCAL.

Audit quantity	Registered value	Interpretation
Deterministic-order bootstrap	$B=20,000$; 10/10 seeds	Raw CI excludes 0 in every audited seed
Cross-seed p-value variation	MC-SE 0.0011; range 0.0144–0.0174	Spans the Holm first-step threshold
Mean smallest p-value	0.01636 vs. $\alpha/3=0.01667$	Knife-edge, not an exact survivor decision
CI-endpoint Monte Carlo drift	MC-SE $\approx 3 \times 10^{-17}$	Bit-identical endpoints from quantized extreme order statistics
Registered family outcome	0 Holm survivors	Retain BORDERLINE_EQUIVOCAL

B INFERENCE CONVENTIONS: ROUNDING, TIES, AND THE TWIN p

Retained count and ties. The selected count at coverage c is $k = \max(1, \text{round}(c \cdot n))$ retained decodes (the top- k by descending score), i.e. *banker’s rounding* of $c \cdot n$ to an integer with a floor at 1 (at $c=0.2, n=974$: $\text{round}(194.8)=195$ decodes retained). Ties in score are resolved by stable (mergesort) index order, so the retained set is deterministic for a fixed event order; this is why the δ_{EL} values are point-identical on the discrete support grid (e.g. $-0.076923 = -15/195$ at $c=0.2$).

Twin p . The *twin p* of an estimate is the two-sided bootstrap p -value against the paired null, $p_{\text{twin}} = 2 \min(\widehat{\Pr}_b[\hat{\delta}_b > 0], \widehat{\Pr}_b[\hat{\delta}_b < 0])$ over the $B=2000$ cluster-bootstrap resamples $\hat{\delta}_b$. It measures direction stability of the estimate on the original (not shuffled) labels, with the independent statistical unit being the decode ($n=974$).

MDE tags. Because MDE_{80} carries one value per scale and coverage, we tag it where ambiguous: $\text{MDE}_{80}^{(\text{AUROC})}$ for the AUROC scale, $\text{MDE}_{80}^{(\text{risk}, c)}$ for the decision-anchor risk@coverage at coverage c , and $\text{MDE}_{80}^{(\text{xblk})}$ for the cross-block anchor. Per-anchor values are in Table 2 (R3) and Table 6 (early/late grid).

C ASSUMPTIONS OF THE AUROC $\rightarrow$ Δ RISK PROJECTION

The first-order projection $\Delta \text{risk} = \Delta A \cdot (1-c) \cdot d$ used in §3 is a scale *projection* of the discrimination estimand into risk units, never a merge of the two estimands. It rests on four explicit assumptions (full derivation in docs/r106_c4_auroc_risk_derivation.md): (i) *monotone score* — selection is by descending score, so the AUC is well defined; (ii) *first-order increment* — Δrisk is a linear-in- ΔA Taylor map, so the reported value is neither a confidence bound nor an equality; (iii) *i.i.d. units* (cluster unit = decode, $n=974$); and (iv) *no calibration assumption* (sorting separability only), so the proxy speaks to the ordering layer and not to the decision-utility layer. Because of (ii)–(iv) the AUROC-implied value is only a first-order point estimate for the decision-readout increment and cannot be separated from it at this bed’s power (“not resolvable”, not “dissociate”). The measured selection estimand remains the between-arm increment of Table 2, and the AUROC-scale $\text{MDE}_{80}^{(\text{AUROC})}$ is never compared directly against a risk-anchor MDE_{80} .

D REGISTERED TABLES MOVED FROM THE MAIN TEXT

Tables 4–8 carry the complete registered values referenced by §2–§4. They are placed here only for the main-text page budget; no value, interval, gate, or caveat differs from the version audited in earlier rounds.

Table 4: Label-permutation zero-effect Type-I calibration of the registered decision rule ($N_{\text{perm}}=1000$ permutations of the decode-level correct label preserving score structure and label marginal; inner multiplicity-preserving cluster bootstrap $B=1000$ over decodes; same 95% CI, twin p , and family Holm(3) gate as the paper; seed 20260825, zero GPU). All values verbatim from `results/r197_labelperm_typeI/r197_labelperm_typeI.json`; the interpretation follows the corrected one-sided fabrication-sanity reading in `results/r197_labelperm_typeI/r198_analysis_conservative.md` (the script’s automated strict-nominal gate demanded a Wilson interval *containing* 0.05, which is not the question asked here). **Takeaway:** every rejection rate lies at or below the nominal 0.05, so the pipeline is *conservative*, not anti-conservative — it cannot fabricate a discrimination signal by over-rejection. Conservatism is disclosed as double-edged: it also makes the R3 non-detection easier to obtain, which is why R3 is reported as a power bound.

Quantity under the zero-effect null	Value	Wilson 95% CI	Reading
Naive rejection, cov= 0.2	0.022	[0.0146, 0.0331]	Entirely below nominal 0.05
Naive rejection, cov= 0.5	0.018	[0.0114, 0.0283]	Entirely below nominal 0.05
Naive rejection, cov= 0.8	0.021	[0.0138, 0.0319]	Entirely below nominal 0.05
Family Holm(3) rejection	0.010	SE 0.0031	FWER controlled below nominal
Pooled commit AUROC	0.7377	—	Discrimination half: single-readout
Pooled all AUROC	0.7100	—	pooled AUROC, observed labels
Observed paired difference	+0.0278	—	Pooled scale, not the intra-block lift
Label-permutation p (2000 shuffles)	0.0005	—	One-sided on difference ; direction-consistent with the cluster-bootstrap $p \approx 6.6 \times 10^{-9}$ for the +0.0144 intra-block lift (different null frames)

Table 5: Aligned paired-contrast engineering: the *same* treatment contrast, `commit - all` on the *same* decode units, evaluated on both scales (verbatim from `results/r203_paired_contrast_commit_vs_all`; zero GPU; multiplicity-preserving cluster bootstrap over decodes). The r12b registered intra-block paired risk grid (Table 2) already uses this contrast; the AUROC paired member is the r8/r203 computation ($B=5000$, seed 4321, the same r8 code path as the freeze). **Takeaway:** on the aligned contrast the paired discrimination contrast is resolvable while every paired selection contrast is `POWER_LIMITED` — the headline conclusion is invariant to the baseline asymmetry of the registered analysis (asymmetric baselines remain disclosed in Figure 1, the Abstract, and Limitation 6). Because both scales now use the same pair of arms, the remaining non-equivalence is purely a units difference, never a claim of dissociation evidence.

Scale	Contrast (paired)	$\hat{\Delta}$	CI95	MDE ₈₀	Gate	Unit
Discrimination	intra-block paired AUROC (<code>commit - all</code>)	+0.0144	[+0.0096, +0.0193]	0.0069	RES	decode
Selection	risk@cov= 0.2 (<code>commit - all</code>)	−0.0058	[−0.018, +0.0058]	0.0170	PL	decode
Selection	risk@cov= 0.5 (<code>commit - all</code>)	−0.0067	[−0.0131, +0.0008]	0.0099	PL	decode
Selection	risk@cov= 0.8 (<code>commit - all</code>)	−0.0008	[−0.0058, +0.0022]	0.0057	PL	decode

Table 6: Early/late decision-anchor grid (δ_{EL}): the window-comparison estimand $\delta_{EL,r}(c) = \text{risk}_{E,r}(c) - \text{risk}_{L,r}(c)$ (early = blocks 0–3, late = blocks 4–7) used by R1, R4, and R5. Under the full-width $\text{MDE}_{80} = 2.8016 \text{ SD}$ rule (uniform, no exemptions; “ $\text{MDE}_{80}/|\hat{\Delta}| \geq 1 \Rightarrow \text{POWER_LIMITED}$ ”), “0 $\notin$ CI” marks whether $\hat{\Delta}$ ’s 95% CI excludes 0. All values verbatim from `r106_c2_anchor_mde_table` (r106; zero-GPU recompute of the frozen r56 full-width MDE_{80} fix; $\text{SD} = \text{cluster-bootstrap SE per anchor}$). A0/A1 controls use the same registered gate on *gain* rather than Δrisk . **Takeaway:** δ_{EL} is power-limited at every anchor (*not* a measured null); the raw $\text{cov} = 0.2$ CI happens to exclude 0 but is not a resolvable R3 increment. The R3 decision quantity is the between-arm increment in Table 2, not this grid. *Uniform MDE80 caliber:* every MDE_{80} here (and in Table 2) is $2.8016 \times$ that *row’s own* cluster-bootstrap SE of the row’s effect of interest (δ_{EL} here; Δ^{intra} there; *gain* for the A0/A1 control rows), all on the risk@coverage scale; rows are therefore not cross-comparable, and the AUROC-scale $\text{MDE}_{80}^{(\text{AUROC})} = 0.0069$ belongs to a *separate* discrimination family that is never compared against any risk-scale row.

Anchor	$\hat{\Delta}$	CI95	MDE_{80}	$\frac{\text{MDE}_{80}}{ \hat{\Delta} }$	0 $\notin$ CI	Gate	Family
commit · 0.2	−0.0769	[−0.144, −0.010]	0.0944	1.227	yes	PL	Early/late δ_{EL} (R1, R4, R5)
commit · 0.5	−0.0267	[−0.058, 0.008]	0.0487	1.824	no	PL	
commit · 0.8	+0.0077	[−0.012, 0.026]	0.0271	3.518	no	PL	
first · 0.2	−0.0769	[−0.133, −0.005]	0.0934	1.215	yes	PL	
first · 0.5	−0.0021	[−0.035, 0.035]	0.0493	24.036	no	PL	
first · 0.8	+0.0013	[−0.021, 0.019]	0.0284	22.136	no	PL	
all · 0.2	−0.0769	[−0.138, −0.005]	0.0953	1.239	yes	PL	
all · 0.5	−0.0246	[−0.055, 0.014]	0.0499	2.025	no	PL	
all · 0.8	−0.0051	[−0.022, 0.018]	0.0284	5.539	no	PL	

Table 7: Equivalence complement and planned-power requirements at the decision anchor. **Takeaway:** the lower-coverage TOST checks are inconclusive on the frozen bed; the current sample is below both planned-power targets.

Check	Anchor	Numeric result	Decision
TOST	cov= 0.2	90% CI [−0.018, +0.004]	Inconclusive at $\delta^* = 1$ pp
TOST	cov= 0.5	90% CI [−0.012, −0.001]	Inconclusive at $\delta^* = 1$ pp
TOST	cov= 0.8	90% CI [−0.005, +0.002]	Equivalent within ± 1 pp
Design power	$\text{MDE}_{80} = 1.2$ pp	$n \approx 1960$	More decodes than the current bed
Design power	$\text{MDE}_{80} = 1.0$ pp	$n \approx 2820$	More decodes than the current bed
Current bed	—	$n = 974$	Below both planned-power regimes

Table 8: Frozen-bed and confidence-readout registry. Numeric schedule and inference settings are grouped here rather than repeated in prose; readout rows hold the source (decode,pos,block) event cell fixed. **Takeaway:** the readouts differ only in within-block aggregation on one frozen event stream.

Registry field	Frozen setting
Bed	LLaDA-8B-Instruct $\times$ GSM8K, <i>free-form exact-match</i> (base acc 0.7053, err 0.2947; <i>not</i> MC 4-way — an earlier label was withdrawn as a misstatement), <code>g3_main/A</code> (registered experiment arm A)
Sample	$n=974$ frozen decodes = 974 distinct problems; one decode per problem
Schedule	8 blocks; block length 32; NFE 64; block b covers steps $8b..8b+7$ (gap 8)
Decode configuration	Temperature $t=0$ (frozen result tag). Remasking policy, generation seed, and prompt template are not recorded in the workspace registry and are therefore not reconstructed.
Answer extraction	A8 canonical free-form numerical judge: <code>#### N > \boxed{} > last-number</code> fallback; absolute tolerance 10^{-6} ; label stored as <code>meta.correct</code>
Inference	Cluster bootstrap $B=2000$ (multiplicity-preserving, fixed seed 20260824); coverage {0.2, 0.5, 0.8}; family-level Holm(3)
commit	last-step conf
first	first-step conf (desaturated, full 8-block coverage)
all	mean conf

Table 9: Frozen R1–R5 hypothesis registry — one auditable table for every registered hypothesis. *Direction* records the pre-registered sign expectation; *Family* the multiplicity family that anchor belongs to (“E/L” = the early/late δ_{EL} family corrected by Holm(3) over $\{0.2, 0.5, 0.8\}$; “AUC” = the separate discrimination-scale family, never merged with E/L risk anchors); *Primary interval* the registered 95% cluster-bootstrap interval; *Verdict* the gate outcome. **Takeaway:** only the R3 discrimination member is above its gate; every risk-scale member is POWER_LIMITED or BORDERLINE, so the load-bearing claim set collapses to “measurable discrimination, power-bound selection”.

ID	Hypothesis	Estimand	Pool	Pre-reg dir	Family	Primary interval	Verdict
R1	Early/late direction-consistent at registered anchor	δ_{EL} (risk, cov .2)	intra-block	< 0	E/L Holm(3)	$[-0.144, -0.010]$	d1 direction-consistent
R2	commit $\neq$ all constructs; gap SNR-compatible	disattenuated reliability bound	same decodes	< 1	—	ub 0.979	distinct constructs; SNR-compatible
R3a	commit discriminative lift vs random	AUROC gain	intra-block	> 0	AUC	$[+0.0096, +0.0193]$	RES (MDE80 0.0069)
R3b	commit selection increment over all	Δ^{intra} risk, cov .2/.5/.8	intra-block	$\neq 0$	decision	all CIs $\supset 0$ (T. 2)	POWER_LIMITED: all anchors reproduced
R4	Early/late survives desaturation	$\delta_{EL, \text{first}}$	8 blocks	same sign	E/L Holm(3)	$[-0.133, -0.005]$	
R5	High-selectivity specificity of gap	$\delta_{EL}(c)$ fine-grid	cov .15..35	neg. gap	E/L Holm(3)	raw CI excludes 0, no Holm survivor	BORDERLINE_EQUIVOCAL